

Engineering Notebook

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Large Scale Design & Analysis of Neural Networks

The engineering notebook is an ongoing document detailing your work completed in each sprint. Each team member will complete their own notebook **independently**. You should list any code you wrote, components you tested, or documentation you wrote. You should also keep track of any meetings you have and what was discussed/worked on during them.

The purpose of the engineering notebook is to provide an unchanging, absolute record of all the work you have done. It records every step and every decision to provide a reference for later decisions. Additionally, it allows others to check your work and progress later. Otherwise, many design and planning decisions may be lost to time.

The engineering notebook should be updated continually throughout the project, not only at the end of each sprint. Each entry must be dated and should always correlate to when the work was completed. Each team member's notebook should be stored within your team's GitHub. Use consistent formatting for the entirety of the notebook, and don't erase previous entries.

The following pages show different examples of how you may choose to format your personal engineering notebook. You do not need to include everything in the examples, but use them as a reference for the amount of effort/size we are expecting to be done.

Date: October 15, 2024

Today I created the Software Requirements Specification and Software Design Documentation documents. I went through the SRR and edited the sections to highlight what needs to be deleted in the end in red, this makes the process of filling it out a lot simpler and makes sure what needs to be deleted is popping out. After that, I wrote the first draft for the Introduction section 1.1 as well as the Distribution list at the top where document control is.

This was done during our meeting in the Lehman building and it was decided that the main editors of these documents would likely be me and Sharif, since for now the only ones working on the main code are Daniel and Simon but that will likely change around as we go sprint to sprint.

I also created the slides file but only added the title & names.

Date: October 18, 2024

I finished the slides today, making sure that they go over things like an overview of our project, what we've done, examples, and what we plan for future sprints. I'm going to wait for a team member to review it before submitting, just in case anything should be added/edited.

Date: October 22, 2024

Today I edited the Software Requirements Specification in section 1.2 for the intended audience and reading suggestions, and section 1.5 (definitions, acronyms, and abbreviations). I double checked with my team about the slides and submitted them before our meeting.

In the meeting, our PhD student John came into the IC 101 to meet with us about our current progress and ask to answer any questions. I personally didn't have questions for him, but my team members asked about a couple things. I think we met for about 15-ish minutes.

Date: October 23, 2024

Today I reviewed my personal code that I was practicing for our project, since I still have some parts of Deep Learning that I don't fully understand quite yet. I'm also adding to the SRS again, these are the parts I filled/finished/started as a part of the first draft of SRS:

- Introduction (1.0 - 1.5) was all finished
- General Description (2.5, 2.6) was started

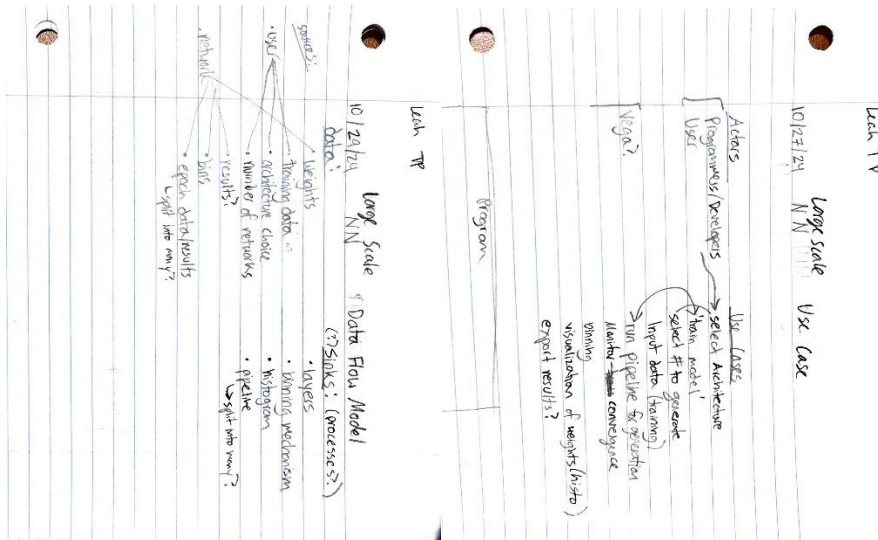
I am also working on the use case model(for 2.3), and after that I plan on making any other diagrams my group thinks is necessary.

Date: October 24, 2024

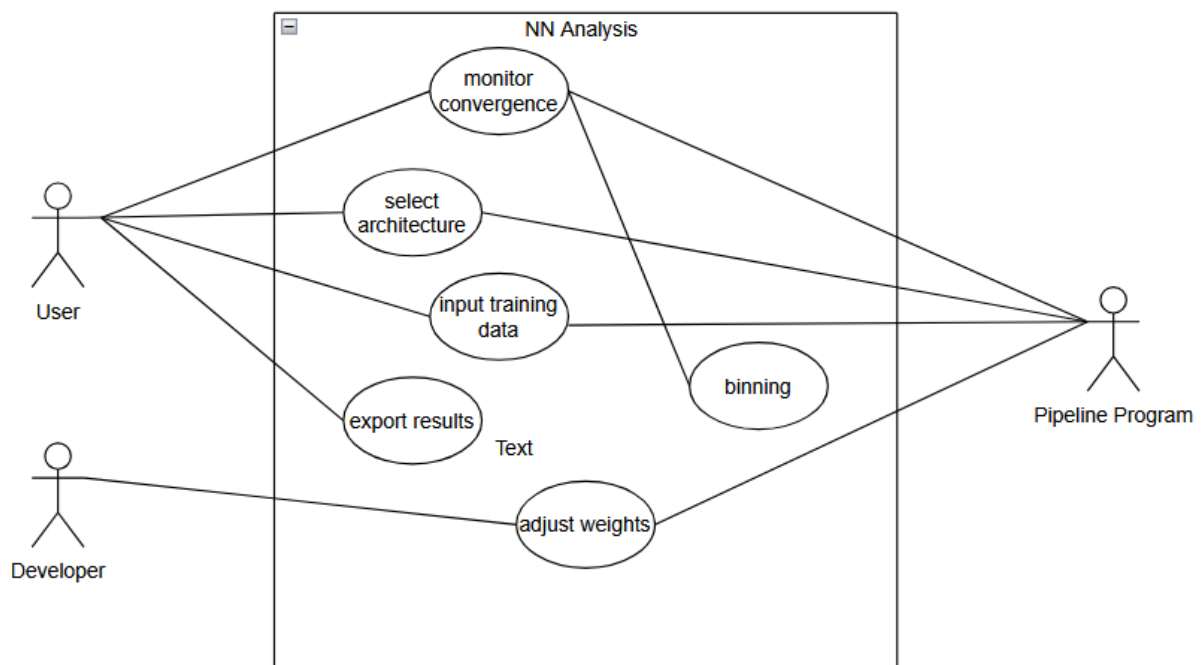
Our group agreed to meet this weekend on Sunday to replace our meeting today due to some members not being able to make it. We also have our submission on canvas due for the notebook and agile so our group created the branch for our notebook and I added points to the tasks I worked on.

Date: October 28th, 2024

Today we were going to finish the SDD but since there was an extension we took class time to work on our individual parts and we had a quick meeting with the PhD student. I reviewed the drafts that I made a couple days ago:



Then I put the Use Case into a rough draft online:

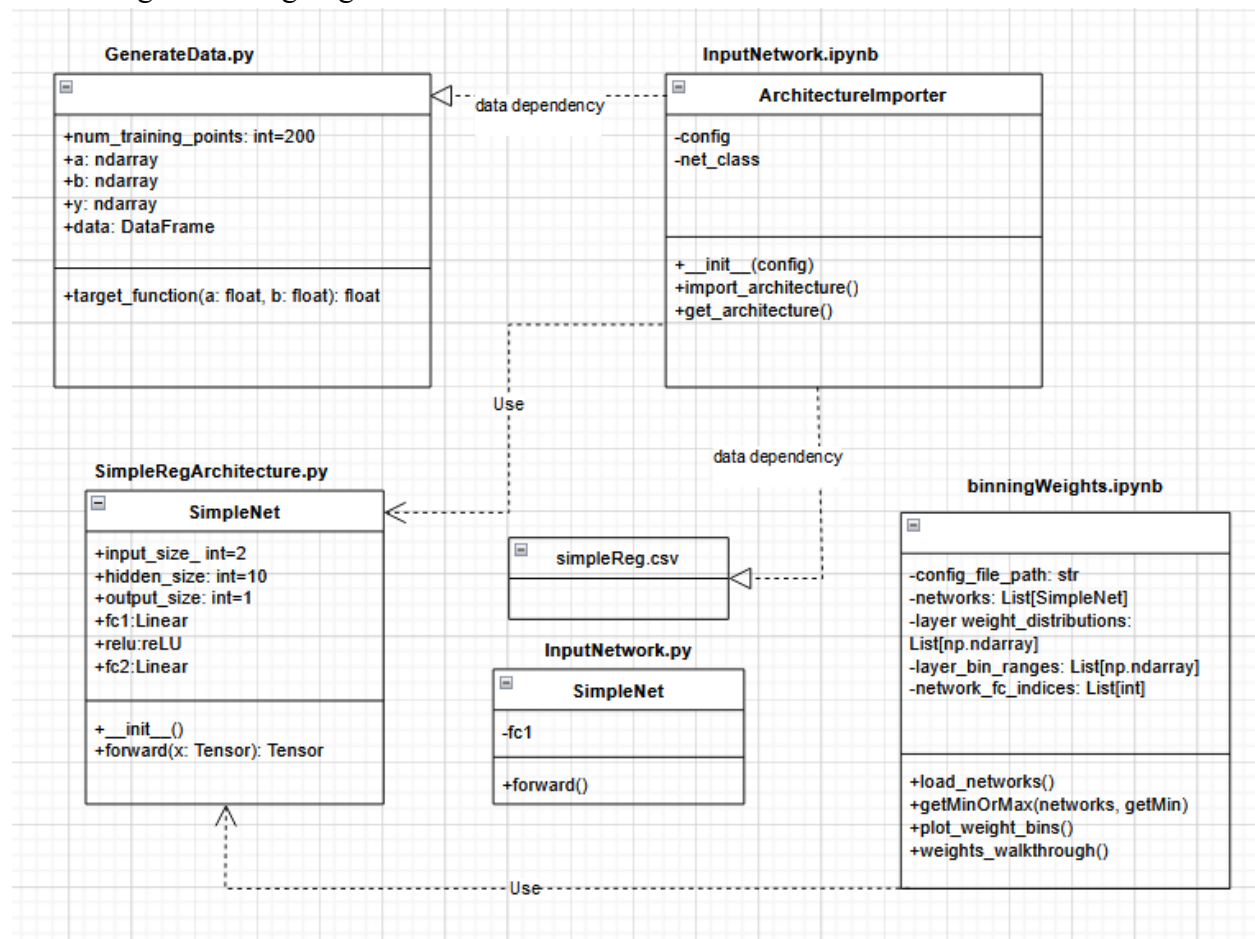


Date: November 12th, 2024

Today we had our presentation, and we also started on the new sprint. I worked with the team to make sure there was a good amount of tasks on the sprint before I started it at midnight. I added the tasks for the Poster draft submission and the documentation for the SDD which we still have to work on.

Date: November 14th, 2024

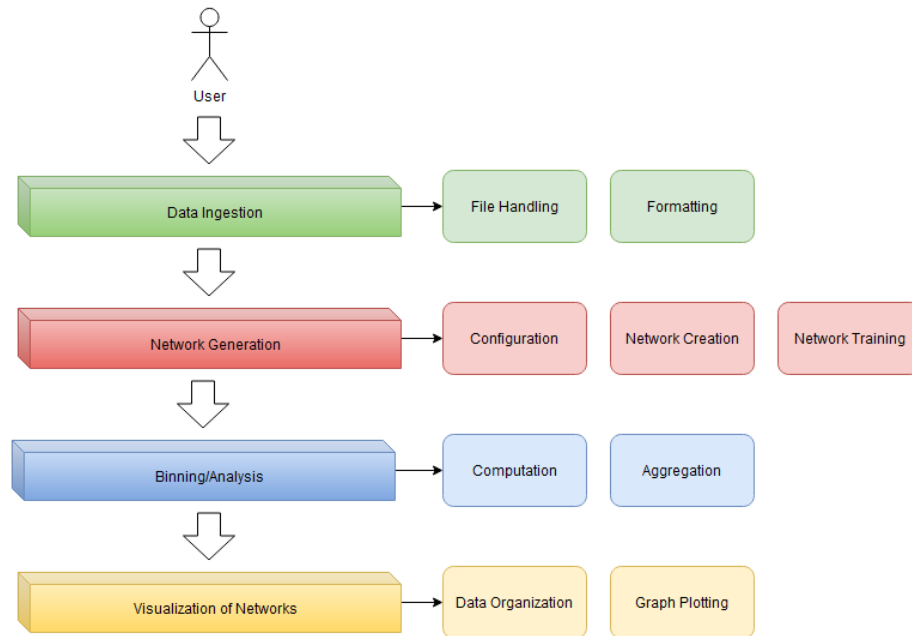
Today I worked on the class models some more so we can add them to the model. I specifically focused on the UML since Daniel said he wanted to work on the data flow model. I first did a scratch-up UML on paper then I made this using the most recent branches of our code from Generating & Binning in git:



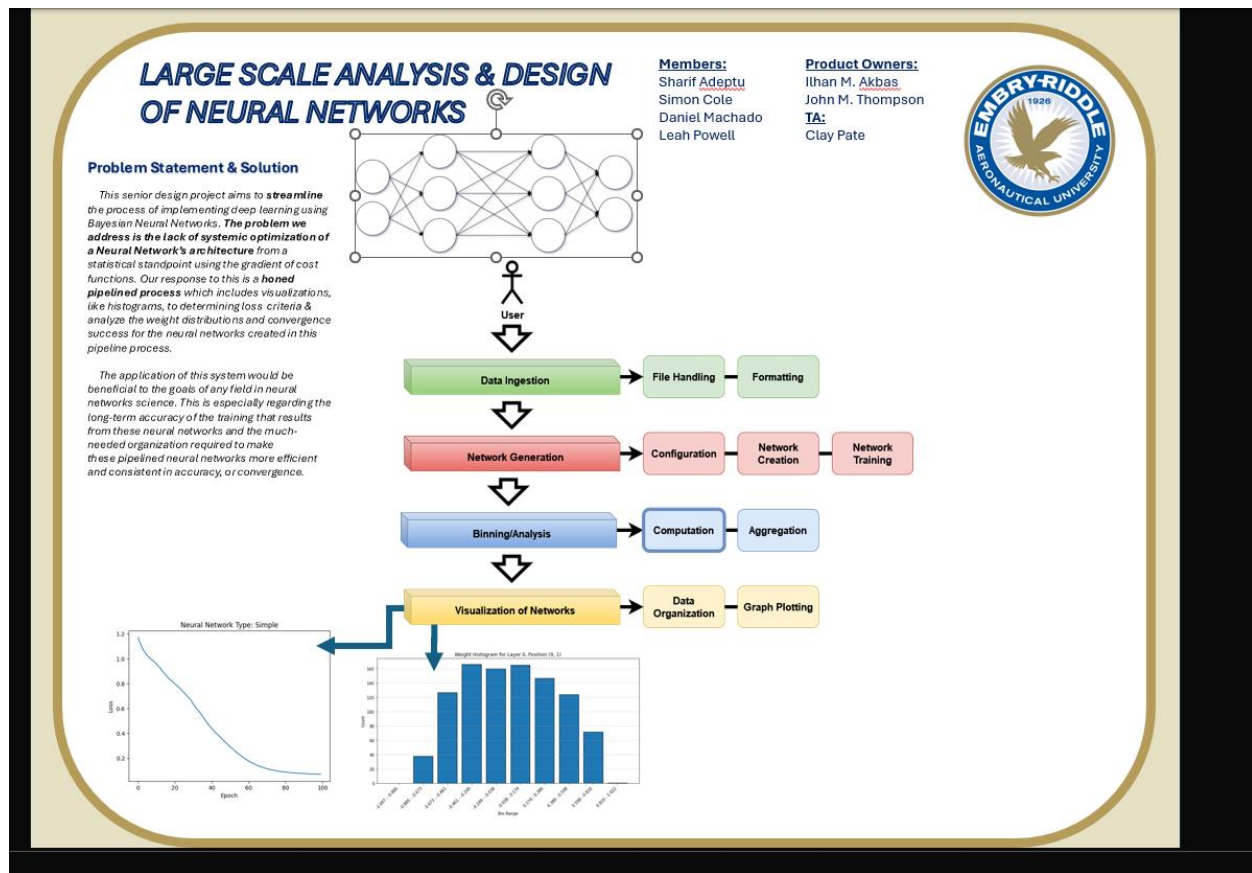
Date: Nov 20, 2024

Today I worked on more of the SDD and added a visual to the poster. I worked with Daniel to complete section 2.2, he wrote the text and I made the diagram for our software architecture. I

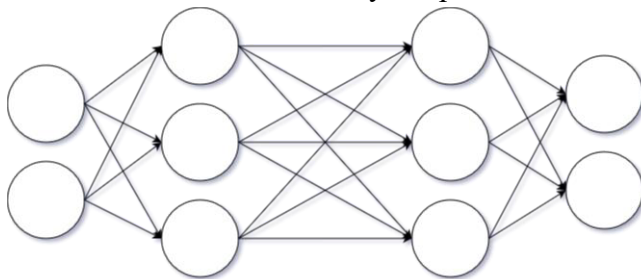
also finished the sections that did not apply to our project by either writing N/A or providing a quick explanation of why that section wasn't relevant. I did a quick sketch of the architecture module before making it in draw.io but it's so messy I won't put it here. Here is the finished 1st draft of the model, though:



After I did this I edited the SDD some more on the rest of the sections that were still empty. There are still one or two sections empty, but I should be able to finish one of them once I finish the low-level collaboration diagram tonight. I also edited and mostly finished the first draft of our poster for presentation day, although I left some white space for my teammates to add anything:



I also made this really simple visual for the poster on Draw.io:

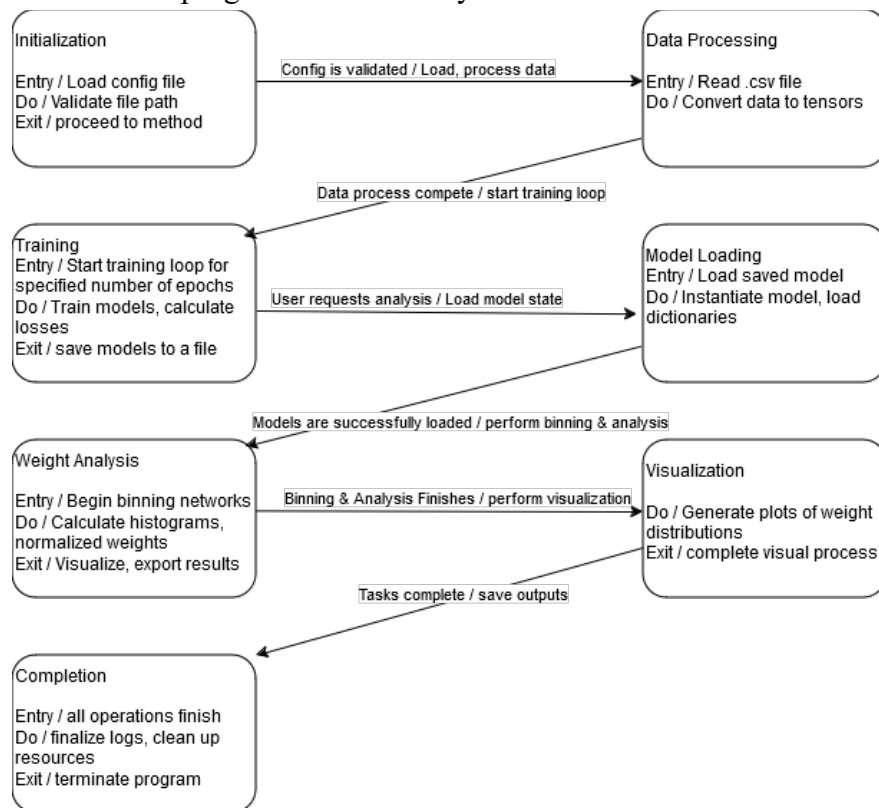


11/26/2024

Today in our class meeting we did not meet with the PhD student. We worked on the SRS and discussed how we should write the requirements with Clay, especially when that section of requirements isn't relevant to our project. I updated the poster and revised it slightly to match the feedback from Clay. However, I will probably keep moving things around in the poster (with group feedback too) for the final submission.

12/1/2024

Today I worked on submitting the SRS for the group. I made sure to finish adding a list of requirements to sections 3.3, 6.1, and 6.2. This also includes adding a State Model Diagram that I made for the program as it currently is to 7.3 in the document.



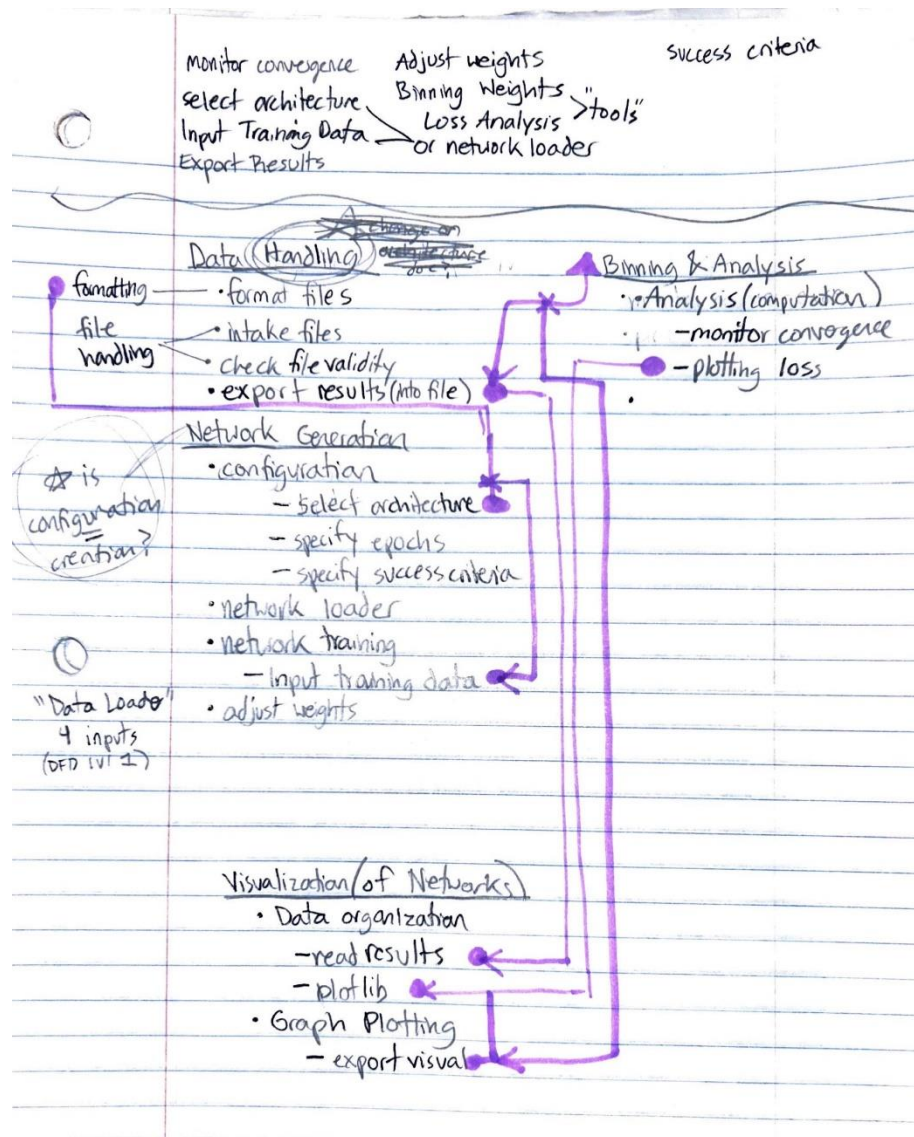
12/3/2024

Today in our meeting we discussed the 10 minute video and the Test plan doc. I made the test plan doc and have started filling out the sections that seem appropriate for us to have, so far I worked on 1.1, 1.3. and 1.7. John (PhD student) also met up with us today and he helped Simon and Daniel understand the code they were working on a bit better and he also was explaining to the group that over the break it might be a good idea to get in touch with him so that we can re-organize the repository to be more accurate for the project, I think he mentioned something was wrong about the branches.

12/5/2024

Today we presented in the LB Atrium and it went really well. We were able to reach a lot of people about our project and I feel like it helped me understand which parts of the project I could improve my understanding of. Last night I worked on the 10 minute presentation video and completed my section that covers the software architecture. While working on this, I thought about some of the disparities between this design and the actual code, especially with how files

are managed. I cleared up some of those misunderstandings with Simon and will be updating the group diagrams to reflect that. I also drew up this very rough relationship sketch about two days ago when I was thinking about how the different modules in our system interact and communicate:



I think that after today's poster session I have a lot of ideas on how to improve our poster for next time:

- Adding labels for diagrams (and make the existing ones bigger) because it was pointed out that the main image had no label so no one knew it was the architecture of our system without having to ask.
- Test out the image quality of all .png files used in the poster (some of it was fuzzy).
- Add clear columns into visual (or use boxes?).
- I'd like to add better coloring to make our diagrams pop more.

- Be sure to include references at the bottom of the poster, including a member's email for contact about the project.

We also have the test plan document due tonight so I am working on that too.